Laboratory Transition and Prototype Protocols for the FTL Program: Soft Scanner Arrays and Hard Carrier Benches

Companion operational manuscript

16 Apr 2026

Abstract

This manuscript operationalizes the first hardware-facing layer of the faster-than-light transport program in prototype and apparatus language. It is downstream of the FTL theory manuscript and the FTL computational manuscript rather than a proof document or a claim of realized faster-than-light transport.

The companion separates the soft route, the hard route, and the first integrated soft-to-hard mission bridge. It defines soft-array readout observables, hard-bench carrier observables, integrated readiness records, protocol families, acceptance gates, and failure modes for prototype work. The present release is restricted to proof-of-regime soft scanner arrays, hard carrier-cycle benches, and an I0-level integrated scheduling target.

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1 Protocol, Boundary, and Laboratory Status Policy

1.1 Purpose of the laboratory companion

This companion translates the pre-engineering transition of the FTL program into apparatus-level practice. Its purpose is to define the first disciplined bridge from theorem-closed route packages to calibratable, controllable, and repeatable prototype protocols.

1.2 Boundary of the manuscript

The present document is downstream to the FTL theory manuscript and the FTL computational manuscript. It is not theorem evidence for the soft closure chain or the hard closure chain, not a claim of unrestricted superluminal transport, and not a finished mission architecture paper.

Shared boundary clause. This companion is downstream to the FTL theory manuscript; it operationalizes the frozen soft and hard route vocabulary in prototype language; and it does not upgrade prototype intent or bench data into theorem evidence.

1.3 Laboratory status labels

The laboratory companion uses a dedicated status system:

- [LA] Laboratory Assumption.
- [LD] Laboratory Definition.
- [LP] Laboratory Protocol.
- [LM] Measured Observable.
- [LR] Laboratory Result.
- [LF] Laboratory Failure Mode.
- [LT] Laboratory Target Claim.

2 Translation Map from Theory to Prototype Practice

2.1 Route objects carried into the prototype layer

Theory object	Prototype counterpart	Measured or controlled channel	What is not claimed
soft branch-locked sector	soft scanner array state	detector contrast, readout timing, memory persistence	no unrestricted remote image theorem from one bench run
finite and hierarchical scanner theorem	finite and multiscale array protocol	pairwise contrast and hierarchy consistency records	no arbitrary-resolution scaling beyond the frozen array family
detector-memory operationalization	persistent readout stack	classicalization error, memory write, readback success	no perfect noiseless detector oracle
hard carrier family	hard carrier bench	carrier window, hold interval, exit verification	no unrestricted faster-than-light object transport claim
one-pole identity package	identity-preserving carrier cycle proxy	pole-survival, identity-fidelity, and return verification channels	no route-level closure from a single cycle
composite binding package	bound-sector bench proxy	binding margin and composite survival readout	no universal composite transport law
mission admissibility and I0	integrated readiness record	soft export, hard initialization, timing compatibility	no complete mission architecture closure

2.2 Frozen first-release package

The present hardware layer freezes three first-release families: a soft finite-array readout family, a hard one-pole carrier-cycle bench family, and an integrated I0 scheduling family in which a soft export chooses a later hard-cycle initialization window.

3 Laboratory Standing Assumptions

3.1 Soft-route assumptions

Assumption 3.1 [LA]. Frozen soft-array package. The soft route is represented at hardware level by finite detector-memory arrays with frozen alphabet size, fixed readout schedule, explicit classicalization window, and repeatable memory-write and readback channels.

3.2 Hard-route assumptions

The hard route is represented by a frozen carrier-cycle bench with four phases: entry, hold, exit, and verification. A hard run is admissible only when carrier observables, pole observables, and verification channels are all logged under one frozen control schedule.

3.3 Integrated-route assumptions

The integrated layer is restricted to I0-level readiness: a soft protocol first exports a validated record, and a hard protocol then uses that record to choose an initialization or verification window. The integrated layer does not yet claim a completed mission stack.

3.4 Control discipline

All route families require explicit logging of timing, control amplitude, detector gain, medium state, and reset conditions. Post hoc reinterpretation of control phases is not admissible.

4 Soft-Route Observables

4.1 Pairwise contrast observable

Definition 4.1 [LM]. Soft pairwise contrast observable. For labels $b \neq c$, the first soft observable is the pairwise contrast

$$C_{bc} = P(\text{read } b \mid b) - P(\text{read } b \mid c),$$

evaluated under the frozen detector and memory protocol.

This is the first prototype-layer surrogate of the scanner theorem. The soft line is unresolved if pairwise contrast collapses at any intended readable scale.

4.2 Classicalization and memory observables

The second soft layer records the tuple

$$W_{\text{soft}} = (\epsilon_{\text{cl}}, m_{\text{pers}}, r_{\text{read}}),$$

where ϵ_{cl} is the classicalization error proxy, m_{pers} is the memory-persistence score over the designated hold window, and r_{read} is the selective readback success rate.

4.3 Hierarchy consistency observable

The multiscale layer records the hierarchy-consistency score

$$\kappa_{\mathrm{hier}} = \text{coarse-to-fine compatibility of the recorded label stack.}$$

This blocks the false inference that a collection of readable local labels is automatically a coherent remote scene.

5 Hard-Route Observables

5.1 Carrier-window and carrier-margin observables

Definition 5.1 [LM]. Carrier-window package. The hard route records the carrier-window package

$$W_{\mathrm{car}} = (\Delta_{\mathrm{gap}}, M_{\mathrm{car}}, t_{\mathrm{hold}}),$$

where Δ_{gap} is the avoided-crossing or branch-separation proxy, M_{car} is the positive carrier-margin proxy, and t_{hold} is the validated carrier hold interval.

5.2 Pole, identity, and binding observables

The next hard layer records

$$W_{\mathrm{id}} = (P_{\mathrm{pole}}, F_{\mathrm{id}}, B_T),$$

where P_{pole} is the pole-survival proxy, F_{id} is the identity-fidelity proxy across one full cycle, and B_T is the composite binding-margin proxy when the bound-sector bench is active.

5.3 Cycle drift observable

The repeatability layer records cumulative cycle drift

$$\delta_{\mathrm{cyc}},$$

which aggregates control drift, margin erosion, and reset failure across repeated cycles. A hard condition is not release-grade when δ_{cyc} grows monotonically across nominally identical runs.

6 Prototype Architectures

6.1 Soft scanner array

The soft apparatus is a finite detector-memory array supporting branch-selective coupling, a classicalization window, memory write, and selective readback. Later multiscale arrays must preserve the cumulative ladder from binary witness to finite scanner to hierarchical scanner.

6.2 Hard carrier bench

The hard apparatus is a carrier-medium bench supporting entry, hold, exit, and verification phases together with pole or resonance tracking, return verification, and identity-preservation checks. Bound-sector runs add composite binding and composite verification channels.

6.3 Integrated scheduler and readiness record

The integrated apparatus layer stores a readiness record containing soft export data, selected hard initialization data, and timing-compatibility metadata. This is the minimal hardware counterpart of the integrated mission record from the theory manuscript.

7 Protocol Families

7.1 Soft finite-array protocol

Protocol 7.1 [LP]. Soft finite-array protocol. Each soft run freezes alphabet size, detector gain, coupling window, classicalization window, memory-write window, and readback rule before the first coupling event. The minimum data product is a contrast table together with classicalization and memory-persistence logs.

7.2 Soft hierarchical protocol

The hierarchical extension repeats the finite-array protocol scale by scale and then records coarse-to-fine compatibility of the exported label stack. This is the first prototype-level proxy for a remote-scene approximation rather than a single readable label.

7.3 Hard carrier-cycle protocol

Protocol 7.2 [LP]. Hard carrier-cycle protocol. Each hard run freezes entry, hold, exit, and verification control phases before the cycle begins. The minimum data product is a carrier-window record, a pole-survival and identity-fidelity record, and a cycle-drift record over repeated runs.

7.4 Integrated I0 protocol

The integrated I0 protocol performs a validated soft export first and then uses that export to choose a hard initialization or verification window. Its purpose is to test timing compatibility and record integrity, not yet to claim a complete soft-hard mission architecture.

7.5 Repetition rule

No release-grade route claim is made from a single run. Each soft, hard, or integrated condition must be repeated three times under the same frozen control and reset state. Any loss of contrast, carrier margin, identity fidelity, or integrated timing compatibility across repetitions downgrades the condition to unresolved.

8 Acceptance Criteria and Failure Modes

8.1 Soft acceptance gate

Target Claim 8.1 [LT]. Soft proof-of-regime target. *The soft route reaches its first hardware target only when a finite or hierarchical scanner array exhibits non-collapsed pairwise contrast, bounded classicalization error, persistent memory readout, and hierarchy consistency across repeated runs.*

8.2 Hard acceptance gate

Target Claim 8.2 [LT]. Hard proof-of-regime target. *The hard route reaches its first hardware target only when a carrier-cycle bench maintains a non-collapsed carrier window, positive carrier and pole-survival margins, validated return identity, and bounded cycle drift across repeated runs.*

8.3 Integrated readiness gate

The integrated I0 target is reached only when the soft export remains readable, the hard cycle remains admissible, and the timing relation between the two remains stable in the same readiness record.

8.4 Failure modes

Failure Mode 8.3 [LF]. Soft contrast collapse. *Pairwise detector contrast fails at one or more intended readable scales.*

Failure Mode 8.4 [LF]. Memory-persistence failure. *The detector shadow does not mature into a stable local record over the required hold window.*

Failure Mode 8.5 [LF]. Hierarchy inconsistency. *Finer labels no longer refine coarser labels into one coherent exported record.*

Failure Mode 8.6 [LF]. Carrier-window collapse. *The hard branch loses a stable positive carrier window or the relevant avoided-crossing gap collapses.*

Failure Mode 8.7 [LF]. Identity-loss failure. *The returned hard-cycle object or sector is not the same verified physical object-sector that entered the carrier phase.*

Failure Mode 8.8 [LF]. Composite-binding failure. *The bound-sector bench loses positive binding margin or composite observability even when constituent carrier phases exist individually.*

Failure Mode 8.9 [LF]. Integrated timing failure. *The soft export and hard initialization windows do not coexist in one stable readiness record.*

9 Reproducibility, Audit Trail, and Non-Claims

9.1 Minimum archival products

Every release-grade condition archives control schedule, detector state, medium state, reset condition, raw trace bundle, derived observable sheet, verdict record, and repetition ledger.

9.2 Operational file crosswalk

The first FTL hardware release is backed by a frozen markdown asset bundle:

- soft protocol: `hardware/protocols/ftl_soft_array_protocol.md`;
- hard protocol: `hardware/protocols/ftl_hard_bench_protocol.md`;
- integrated readiness protocol: `hardware/protocols/ftl_i0_readiness_protocol.md`;
- run sheet: `hardware/protocols/ftl_route_run_sheet_template.md`;
- baseline BOM: `hardware/bom/ftl_i0_baseline_bom.md`;

- threshold package: `hardware/calibration/ftl_route_threshold_package.md`;
- anchor library: `hardware/calibration/ftl_route_anchor_library.md`;
- geometry note: `hardware/drawings/ftl_soft_hard_geometry_spec.md`.

9.3 Forward dependency order

The prototype dependency order is

$$\begin{aligned}&\text{soft readable label} \to \text{soft persistent record} \to \text{soft hierarchy consistency}\\ &\to \text{hard carrier window} \to \text{hard identity-preserving cycle} \to \text{integrated readiness.}\end{aligned}$$

This order is strict and mirrors the transition from route closure to pre-engineering closure in the theory manuscript.

9.4 What is not claimed

The present laboratory companion does not claim unrestricted faster-than-light signaling or transport, theorem evidence for the soft or hard route theorems, completed material embedding, or a finished mission-ready architecture.